

MATTHEW DIBBLE

Highly motivated and passionate graphics engineer with 4+ years experience in AAA games

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SKILLS

Languages	C++, Rust, C, HLSL, Slang, Python
Tech	DX12, Vulkan, Metal, Agc, PIX, Razor GPU, Nsight, RenderDoc, Visual Studio, Perforce, Git
Skillset	GPU ray tracing, GPU debugging & optimization, multithreaded programming, large codebases
Platforms	PlayStation, Xbox, Windows, macOS

EXPERIENCE

Activision	September 2022 - Present
Software Engineer, Graphics	Toronto, Canada

- Primary task: Rewriting Call of Duty's internal tools renderer that powers baking and level editing workflows
- Added **real time GPU path tracing** support to the renderer to greatly enhance artist workflows
- Migrated the **secondary specular baking** pipeline to use GPU path tracing, massively improving bake times
- Transitioned the renderer to a **bindless architecture** for materials and geometry, as needed by ray tracing
- Added support for **visibility buffer rendering** to decrease GPU render time for dense meshes such as foliage
- Added **volumetric lighting**, screen space shadows/reflection/refraction, GTAO, and subsurface scattering
- Added **water rendering** support with surface wave particle simulation and underwater volumetric lighting
- Revised post processing pipeline by adding **HDR** support, **DLSS**, ACES 2.0, DoF, and temporal accumulation
- Converted the renderer to use a **render graph** to automate dependency and barrier management
- Credits: *COD: Modern Warfare III*, *COD: Black Ops 6*, *COD: Black Ops 7*, *COD: Modern Warfare 4*

PROJECTS

Modern Physically Based Renderer (Raster and Ray Traced) — C++, Vulkan, Slang	Source
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- Scalable **raster and ray tracing** PBR renderer adhering to **modern rendering principles**
- Everything is **bindless** and utilizing modern Vulkan features (BDA, push constants, one static descriptor set)
- Integrated support for TAA, bloom, soft shadows (PCSS), auto-exposure, GT7 tonemapping, HDR support
- Raster: **GPU-driven** draw calls/LOD selection/culling, vertex pulling (no explicit vertex streams or IA)
- Path Tracer: **Importance sampling**, temporal accumulation, Russian roulette, shared material evaluation

Vulkan and Metal RHI — C++, Metal, Vulkan	Source
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- Scalable RHI containing backends for **Metal and Vulkan**, scaling from mobile to high-end desktop GPUs
- Includes a **mature and featured PBR GLTF viewer** as a reference demo application

Nintendo Entertainment System (NES) Emulator — Rust, SDL2	Source
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- Cycle-accurate emulator of the *NES* that takes advantage of **Rust**'s unique properties
- Degree of accuracy allows for play of *Super Mario Bros.*, *The Legend of Zelda*, and much more

EDUCATION

Bachelor of Science in Computer Science, Wilfrid Laurier University	September 2018 - August 2023
Bachelor of Business Administration, Wilfrid Laurier University	Waterloo, Canada

- Dual Degree — 3.9/4.0 GPA
- Teaching Assistant & Grader, Data Structures II